Industrial Internship Report on

”Online voting system using face recognition”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner Uni Converge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was 'Online Voting System Using Face Recognition'. The system verifies voters using facial recognition before allowing them to vote, improving security and preventing duplicate voting. The project was developed using Python, OpenCV, face recognition techniques, HTML, CSS and SQLite. During the internship, I gained practical experience in software development, testing, GitHub usage, documentation and problem solving. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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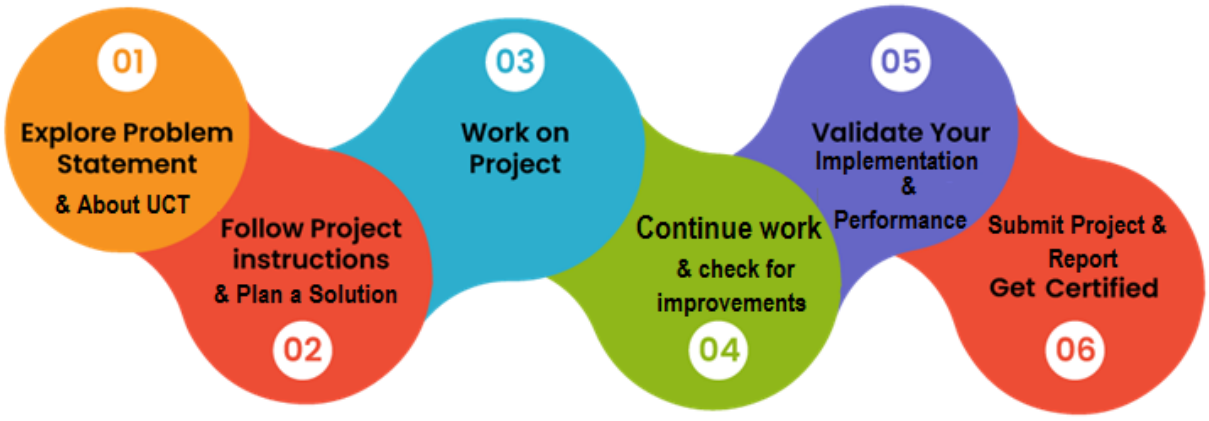
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1 Preface

This report summarizes my six-week internship experience.

The internship helped me understand real- world software development and improve my technical and communication skills. I sincerely thank Upskill Campus, The IoT Academy, UniConverge Technologies and my college for this opportunity.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



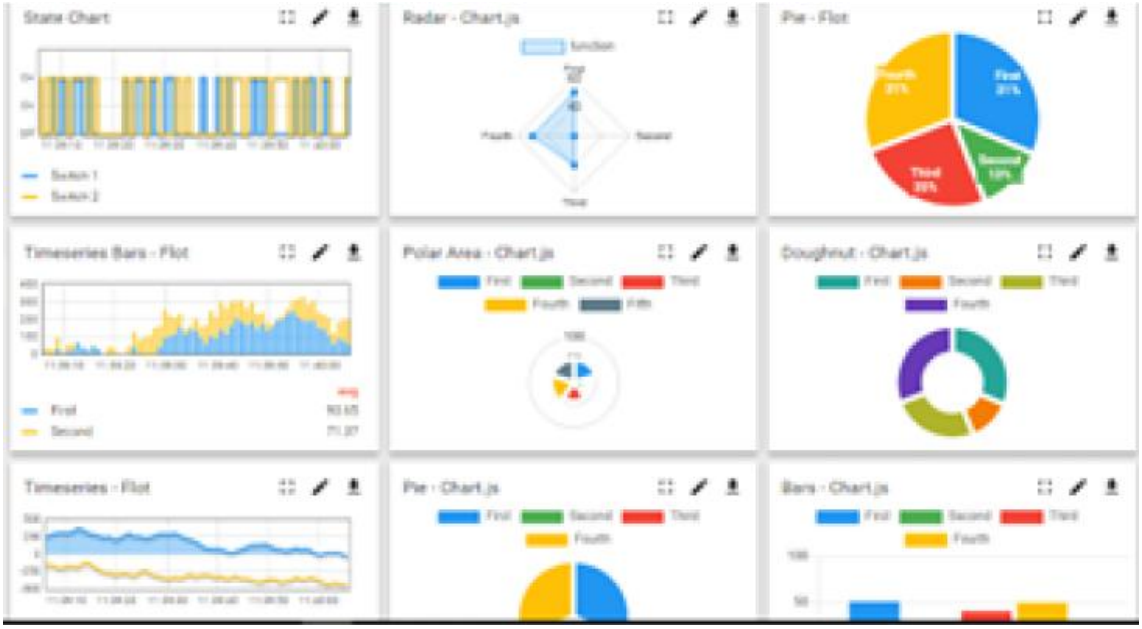
i. UCT IoT Platform (UCT Insight)

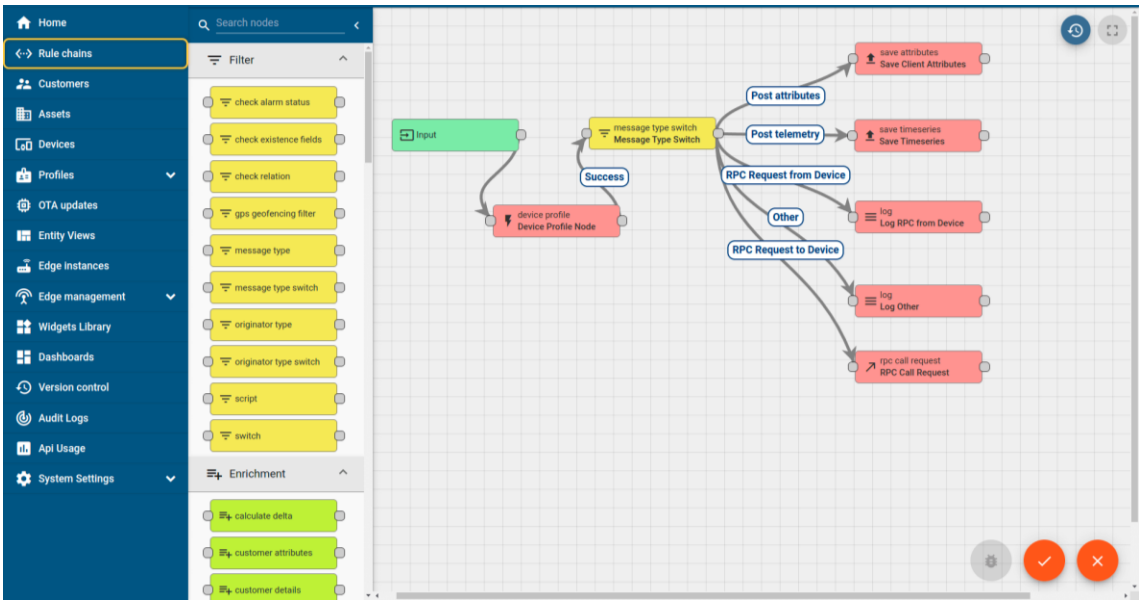
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

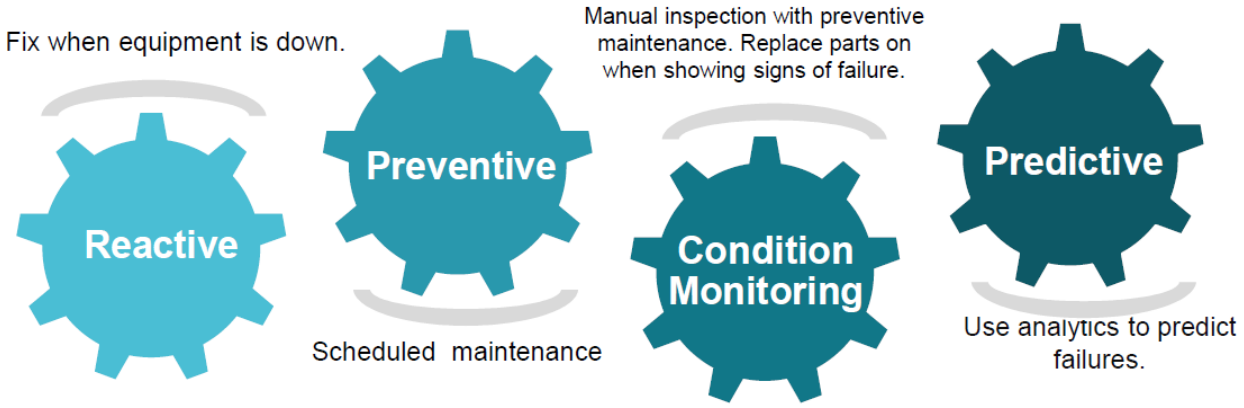


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

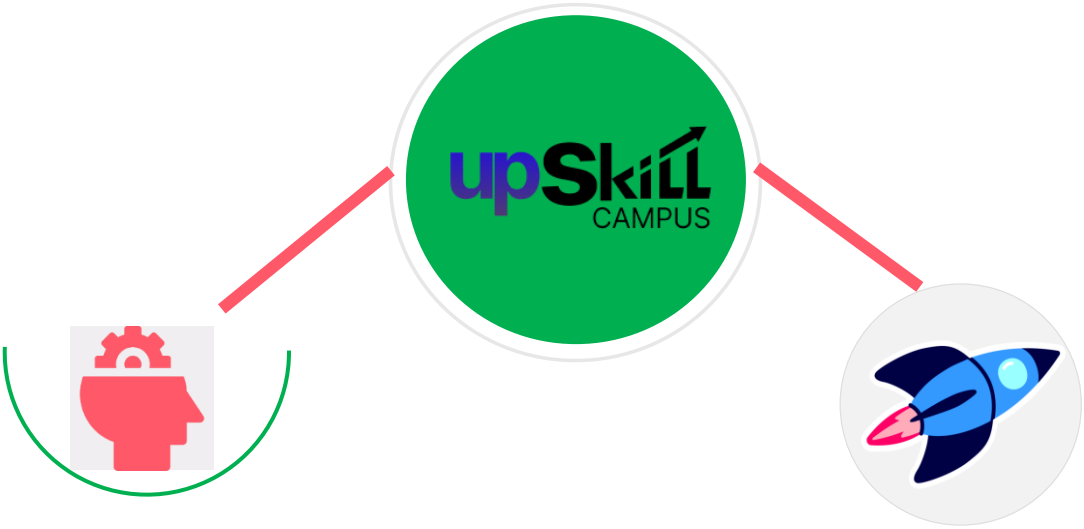
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

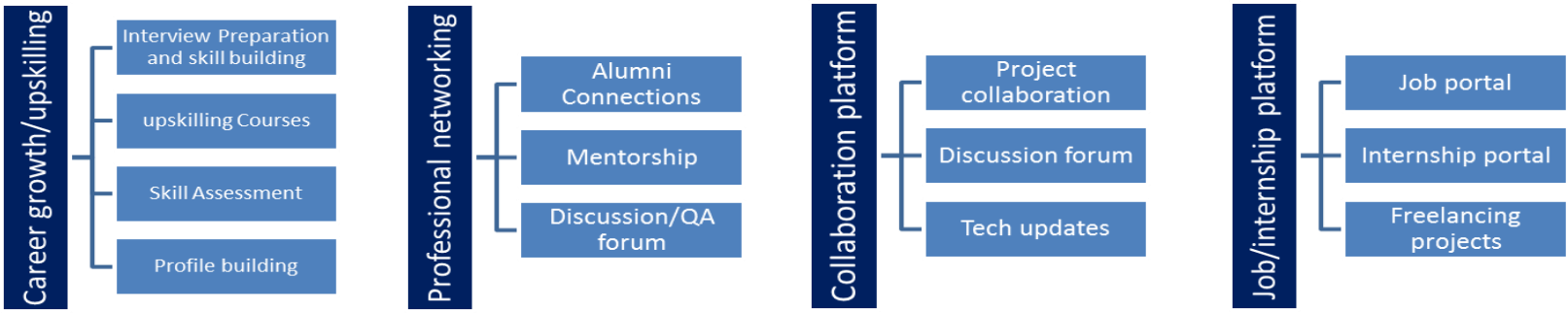
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] https://docs.opencv.org/

[2] https://docs.python.org/3/

[3] https://docs.github.com/

2.6 Glossary

Terms	Acronym
AI	Artificial Intelligence
ML	Machine Learning
CV	Computer Vision
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets

3 Problem Statement

Traditional voting systems are time-consuming and vulnerable to identity fraud. The aim of this project is to authenticate voters using face recognition and ensure that each registered voter can vote only once.

4 Existing and Proposed solution

Existing systems rely on manual verification. The proposed solution uses facial recognition for authentication, providing faster, secure and transparent voting.

4.1 Code submission (Github link)

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model

Workflow: Registration → Face Capture → Face Verification → Login → Cast Vote → Store Vote → Display Success.

5.1 High Level Diagram

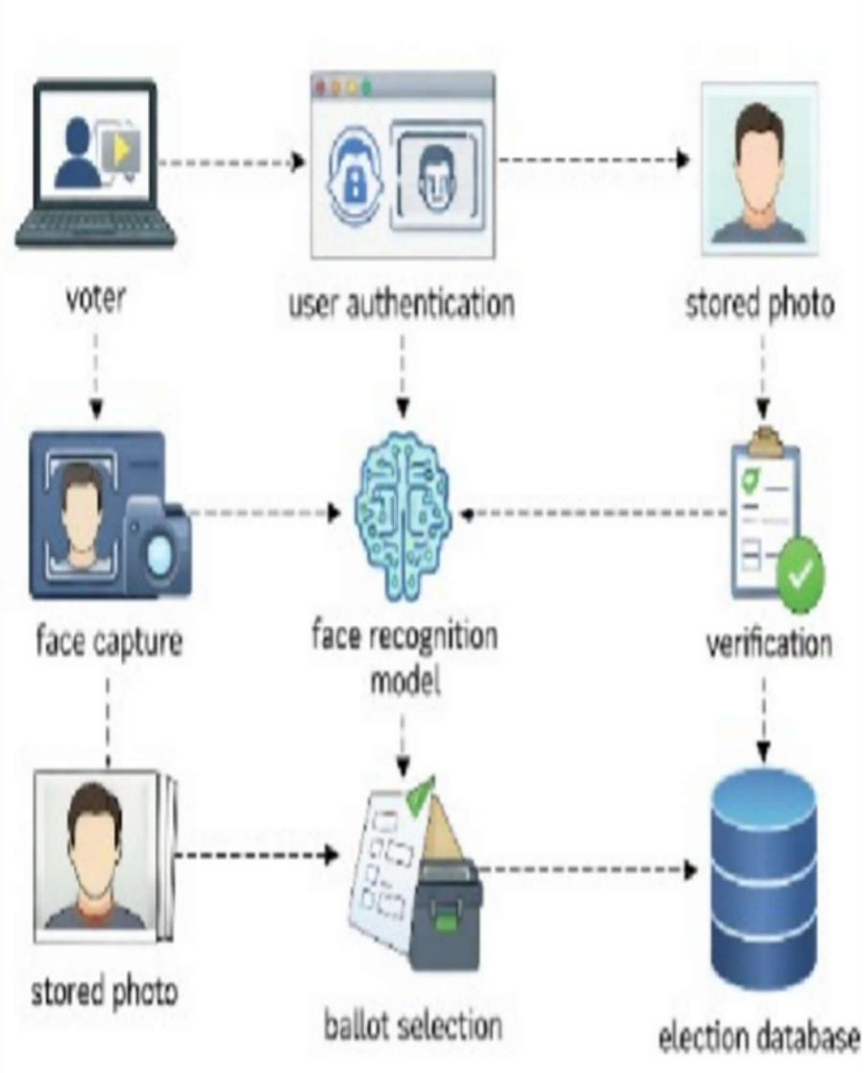
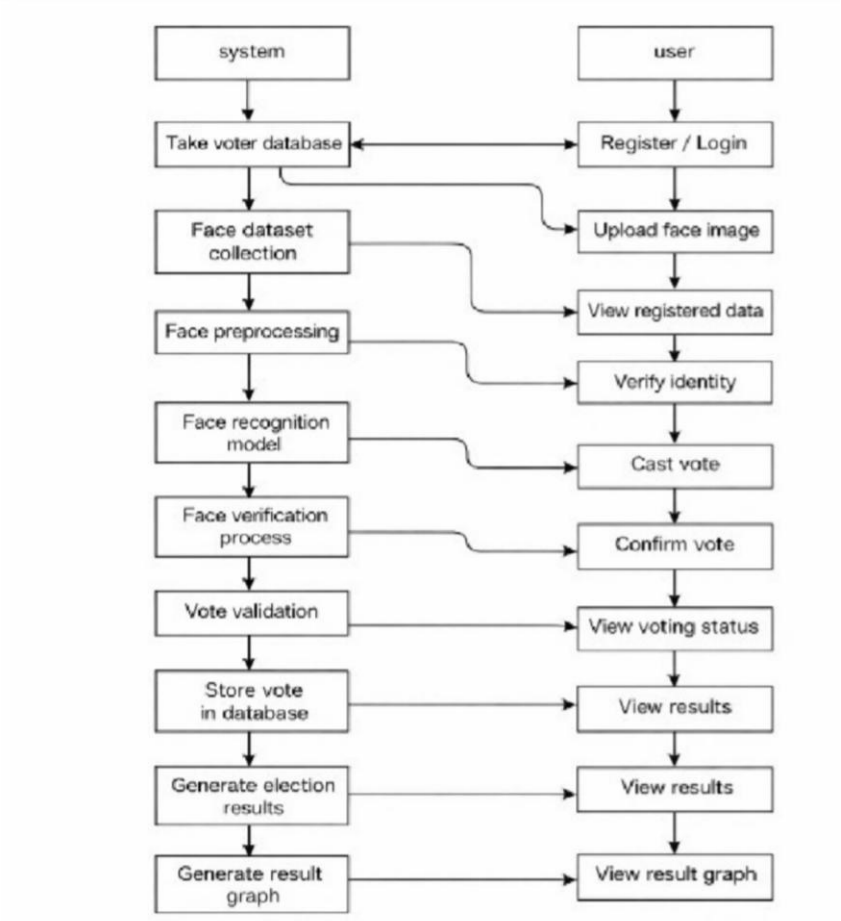


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram



6 Performance Test

The performance of the proposed Online Voting System Using Face Recognition was evaluated based on authentication accuracy, response time, reliability, database operations, and security. The system was tested under different scenarios to ensure that only authenticated users could cast their votes and that each voter was allowed to vote only once.

The major constraints considered during testing were:

- Face recognition accuracy under different lighting conditions.
- Processing time for face verification.
- Prevention of duplicate voting.
- Secure storage of voter information and voting records.
- Efficient database performance during registration and voting.

The system was designed to overcome these constraints by using face recognition for identity verification, maintaining a secure voter database, validating each vote before submission, and preventing duplicate voting through voter authentication.

6.1 Test Plan/ Test Cases

Test Description	Input	Expected Output	Result
Face recognition During login	Voter shows Registered face to camera	System should recognize the voter and allow access to voting page	Pass
Candidate Display List	Login successful	List of candidates should be displayed to the voter	Pass
Duplicate Vote Prevention	Same voter tries to vote again	System should display “You have already Voted”	Pass
System Security Check	Unauthorized access attempt	System should deny Access and show error message	Pass

6.2 Test Procedure

1. Register a new voter with personal details and face image.
2. Log in using the registered credentials.
3. Capture the live face image using the webcam.
4. Compare the captured image with the stored facial data.

5. Verify the identity of the voter.
6. Allow the voter to cast a vote if authentication is successful.
7. Store the vote securely in the database.
8. Prevent duplicate voting attempts.
9. Generate and display the election results after voting is completed.

6.3 Performance Outcome

- The face recognition module successfully authenticated registered users.
- Unauthorized users were denied access to the voting system.
- Duplicate voting attempts were successfully prevented.
- Voter information and voting records were stored securely in the database.
- The voting process was completed quickly with minimal response time.
- The election results were generated accurately after successful vote submission.
- The overall system performed reliably and provided a secure, transparent, and user-friendly voting experience.

7 My learnings

During this internship, I gained valuable practical knowledge and hands-on experience in developing an Online Voting System Using Face Recognition. This project helped me understand the complete software development process, from requirement analysis to implementation, testing, and documentation.

Through this project, I learned:

- Improved my Python programming skills by developing a real-world application.
- Learned to use OpenCV for face detection and image processing.
- Understood the implementation of Face Recognition for user authentication.
- Gained experience in designing a secure online voting system.
- Learned to work with databases (SQLite/MySQL) for storing voter information and voting records.
- Improved my knowledge of HTML, CSS, and Flask for developing the user interface.
- Learned to integrate different software modules into a complete working application.
- Understood the importance of authentication, data security, and preventing duplicate voting.
- Improved my debugging, testing, and problem-solving skills.
- Learned to use Git and GitHub for project management and version control.
- Developed technical documentation and report-writing skills.
- Enhanced my communication, time management, and project planning abilities while completing the internship.

8 Future work scope

In the future, the system can be enhanced by adding more advanced security features. The system can be integrated with national identity databases for better voter verification.

Mobile application support can also be added so that voters can vote using their smartphones. Additional technologies such as fingerprint authentication and blockchain-based security can be implemented to further improve the reliability and security of the voting system.

The system can also be expanded to support large-scale elections and handle a large number of voters efficiently.